

MG Engine

Minority-Game Systematic Trading Research Platform — equity-index futures

Self-directed research project

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Overview

A deterministic, replay-safe, bar-by-bar trading engine for equity-index futures, built in R (approx. 60,000 lines) and grounded in Minority Game statistical mechanics (Challet, Marsili, Zhang, Coolen). The same configuration on the same data reproduces the same output, bar for bar.

Architecture

- Seven-layer per-bar pipeline: MG signal $\rightarrow$ edge validation $\rightarrow$ risk sizing $\rightarrow$ governance $\rightarrow$ per-frequency intent $\rightarrow$ execution latch $\rightarrow$ fill, under a binding $decision(t) \rightarrow execution(t+1)$ timing law.
- Strict single-writer state — every surface has exactly one owning module; a projection / selection / commit phase split per layer makes all data flow auditable.
- Multi-frequency consensus on prime-number timeframes (1m execution carrier + 7m / 13m / 19m voters) to avoid trader-cluster boundaries.
- approx. 149 per-bar attribution surfaces tracing every decision from raw signal to realized fill.

Theoretical basis

- Latent agent ecology: N agents per timeframe, each holding S strategies that map a market-history index to an action, with virtual-score learning and grand-canonical abstention.
- Per-bar order parameters: predictability H , volatility σ^2 , real-vs-fake payoff Δ , strategy score gap, crowd concentration, frozen fraction.
- Implements mechanics from Coolen (cavity / generating-functional analysis), De Martino (mix-game), Marsili (producer–speculator), Sasidevan (stochastic strategies), Kalinowski (multi-memory) and Ferreira–Marsili (autocorrelation-adaptive payoff).

Validation & rigor

- Cross-anchor study over 20 anchors spanning 2022–2024 regimes (trend up/down, central-bank windows, stress).
- Walk-forward in-/out-of-sample split; bootstrap confidence intervals ($B = 1000$); Monte-Carlo trade-order shuffles ($M = 10000$); vol-scaled (Almgren–Chriss-style) cost model.
- Self-falsification discipline: a multi-state regime machine was designed, implemented behind audit flags, verified across anchors, and reverted when it produced no net lift — never curve-fit to the sample.

Results — honest

- Research-grade: a measurable gross signal exists, but the net edge is marginal and information-limited under OHLC inputs — consistent with the Glosten–Milgrom information ceiling; a volume (OHLCV) extension was wired in, cross-anchor-tested, and reverted (no net lift — the ceiling held under OHLCV).
- The deliverable is reproducible methodology and engineering integrity — deterministic replay, full attribution, honest negative results — not an inflated P&L claim.

Engineering & stack

- R; the architecture (single-writer, deterministic replay, strict $t+1$ timing) ports cleanly to Python / C++.
- “The logic carries the bot” — no curve-fit thresholds, no sample-tuned weights, no side-disable filters; every parameter is justifiable independently of the current sample.

Deterministic · fully attributed · honestly validated.